



Behind Closed Doors: Physical Red Team Tactics

Firat Acar, Moritz Thomas – August 2025

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Content

- About us
- Background (light)
- Stories
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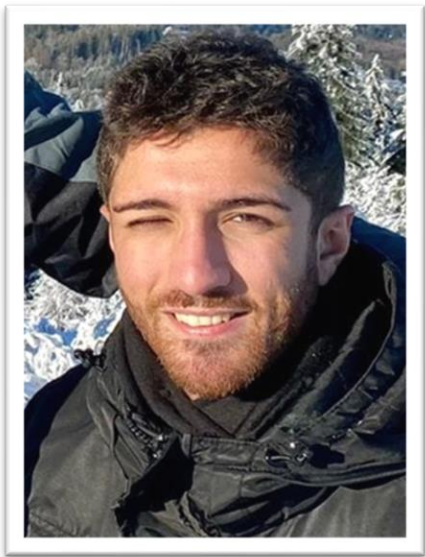


Disclaimer:

Story details (names, locations, sectors, dates, ...) are anonymized.

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About us



Firat Acar

Red Teamer

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About us



Moritz Thomas

Red Teamer, R&D

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About NVISO

OUR COMPANY

Since 2013, NVISO has been offering cybersecurity services across Europe as a dedicated cybersecurity firm with **more than 270 experts**. We have established branches in Belgium, Germany, Austria, and Greece.

Our mission is to **protect the foundations of the European community** against cyber attacks. Our TLPT tests are an example of this.

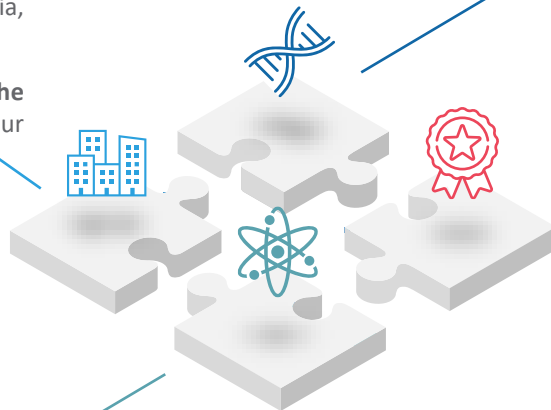
OUR RESEARCH

We invest **10% of our annual revenue** in researching new security techniques and developing new solutions.

We share our findings through **blog posts** and regularly give presentations at **leading cybersecurity conferences**.

 @NVISOsecurity
@NVISO_Labs

 blog.nviso.eu



OUR JOB

Prevent attacks before they occur by testing applications, environments and organizations, and hardening business processes.

Detect adversaries during their attack executions by supporting our clients' SOC or bringing our managed SOC, providing visibility and agency to defenders.

Respond to observed and ongoing attacks with our DFIR team, minimizing fallout and recovering regular operation.

OUR OFFER

NVISO can support you throughout the entire lifecycle of **cybersecurity incidents**. We have extensive experience in providing cybersecurity services for the **finance, government, defence, and technology** sectors.



Background ⚡

Red Team vs Pentest



Security testing of different **technologies / applications**, e.g.
Web, Mobile, WIFI, Cloud,
Internal / External Network



Attack simulation against a **company** with objectives, stealth,
wide scope (including **people**),
phishing, physical breach

Physical Methodology



Red Team – Who is Who?



Red Team

- Executes an attack scenario stealthily
- Has to find their way in, either via physical or cyber intrusion
- Tries to reach one or more objectives

White Team

- Fully aware of the operation
- Involved in all planning
- Connection between Red and Blue Team
- Frequent and direct communication with Red Team

Blue Team

- Usually not (fully) aware of the attack
- Supposed to treat incidents as real and investigate
- Make the Red Teamer's life difficult



Story time!



Industrial Plant

Industrial Plant

Objectives:

- Gain physical access
- Gain access to servers
- Plant rogue device

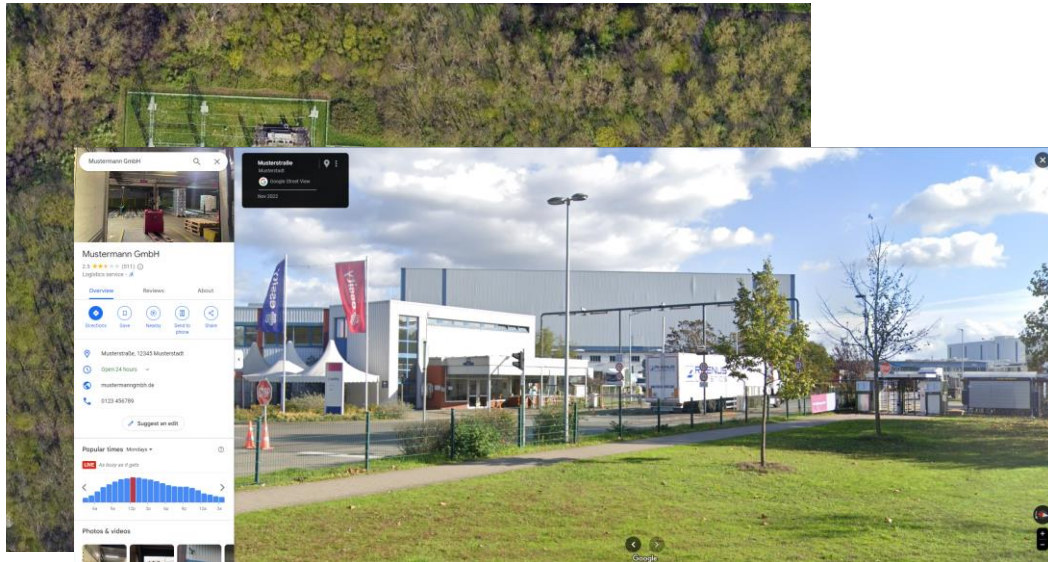


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access



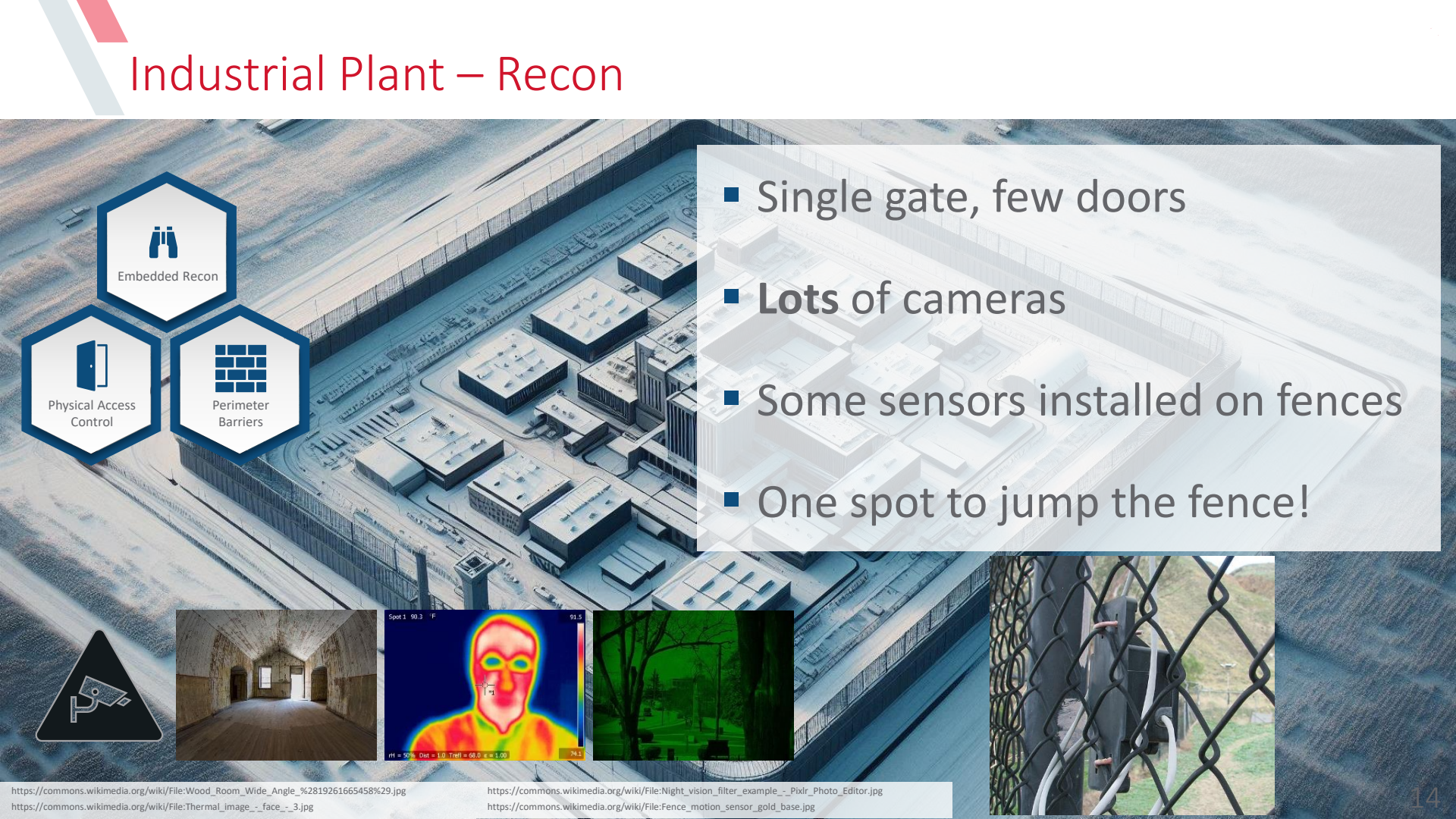
Hardware
Additions

Industrial Plant – Recon



- Remote location
- Extensive area
- Closed to public

Industrial Plant – Recon



- Single gate, few doors
- **Lots** of cameras
- Some sensors installed on fences
- One spot to jump the fence!

Embedded Recon

Physical Access Control

Perimeter Barriers



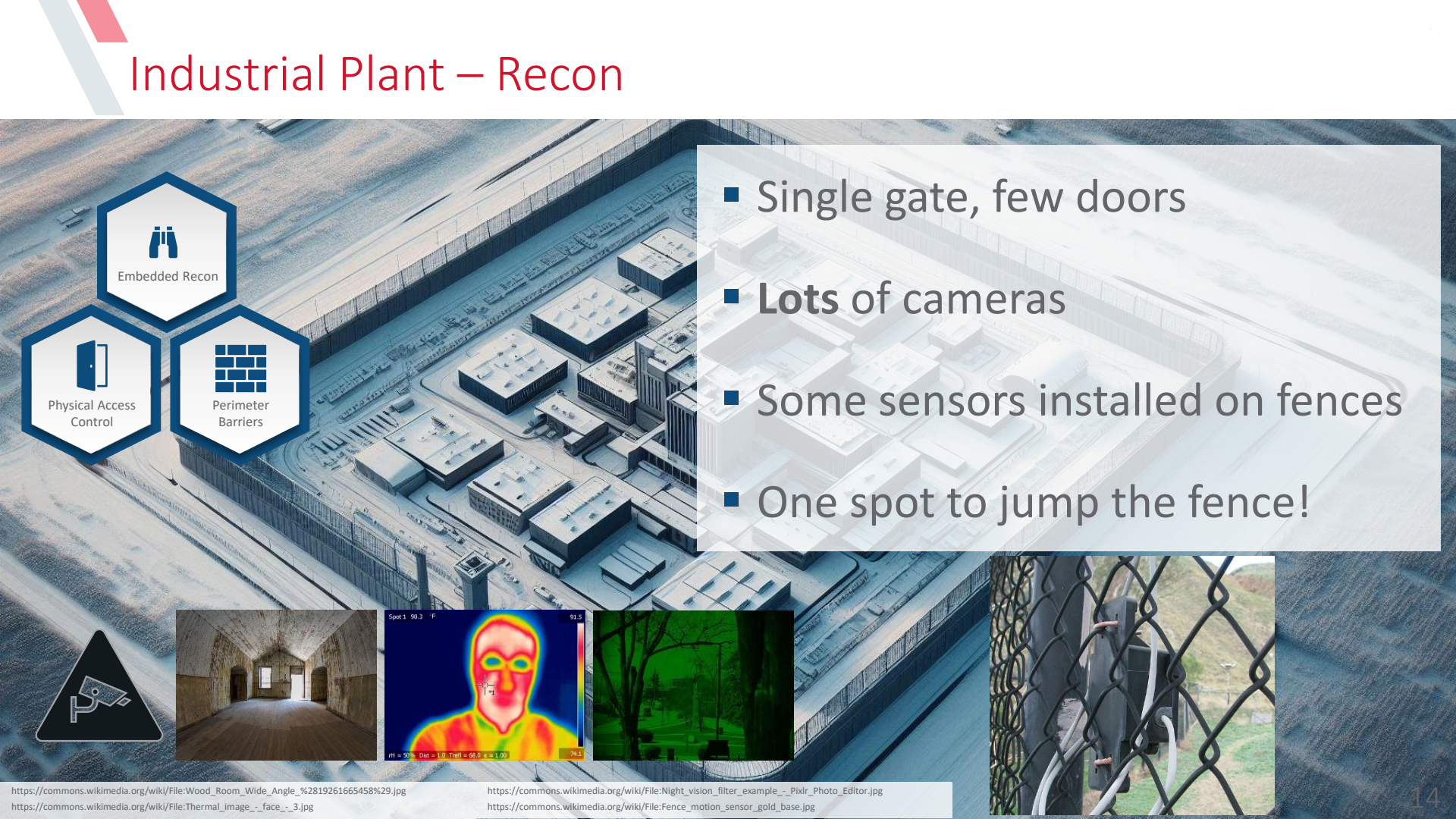
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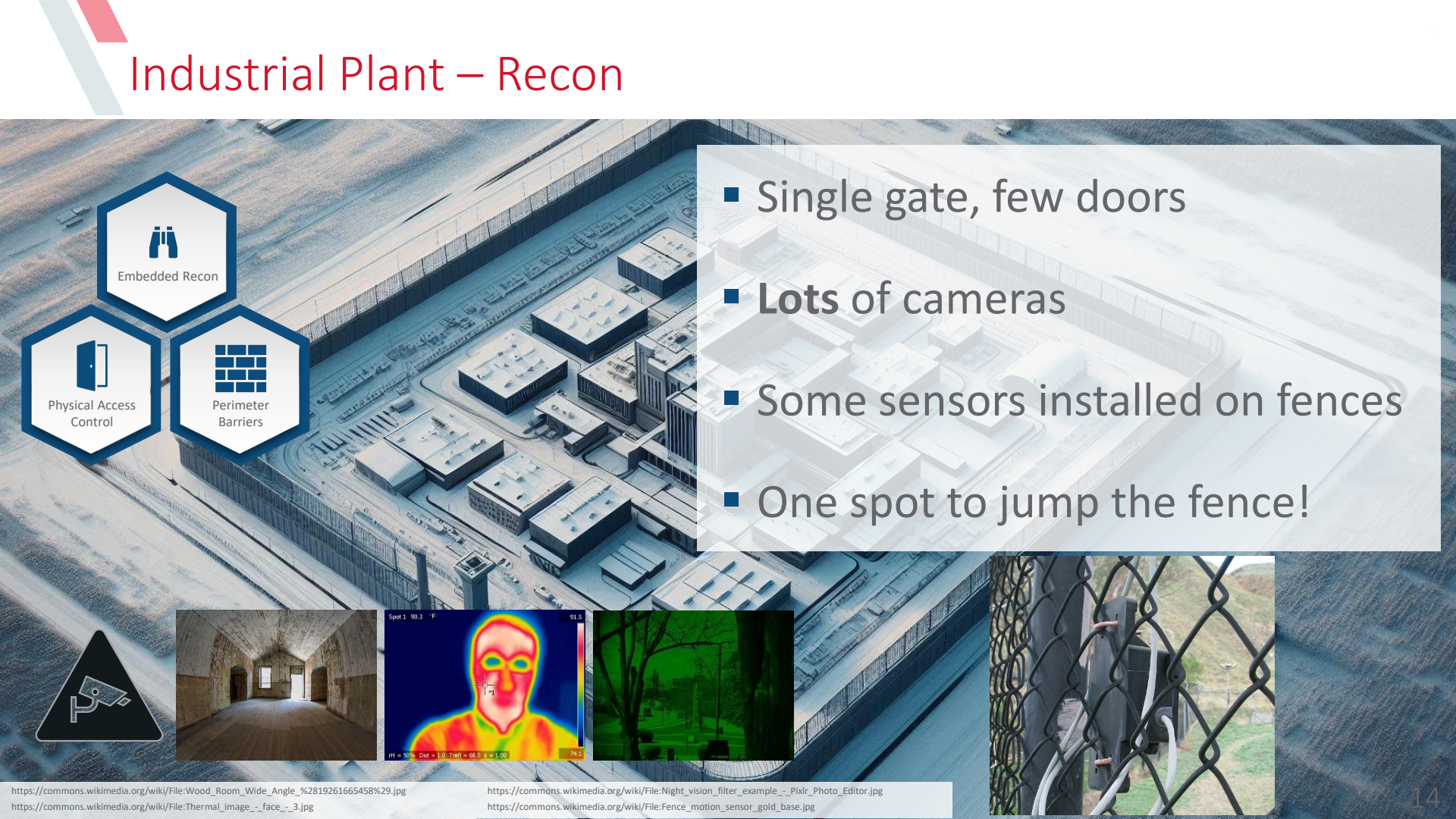
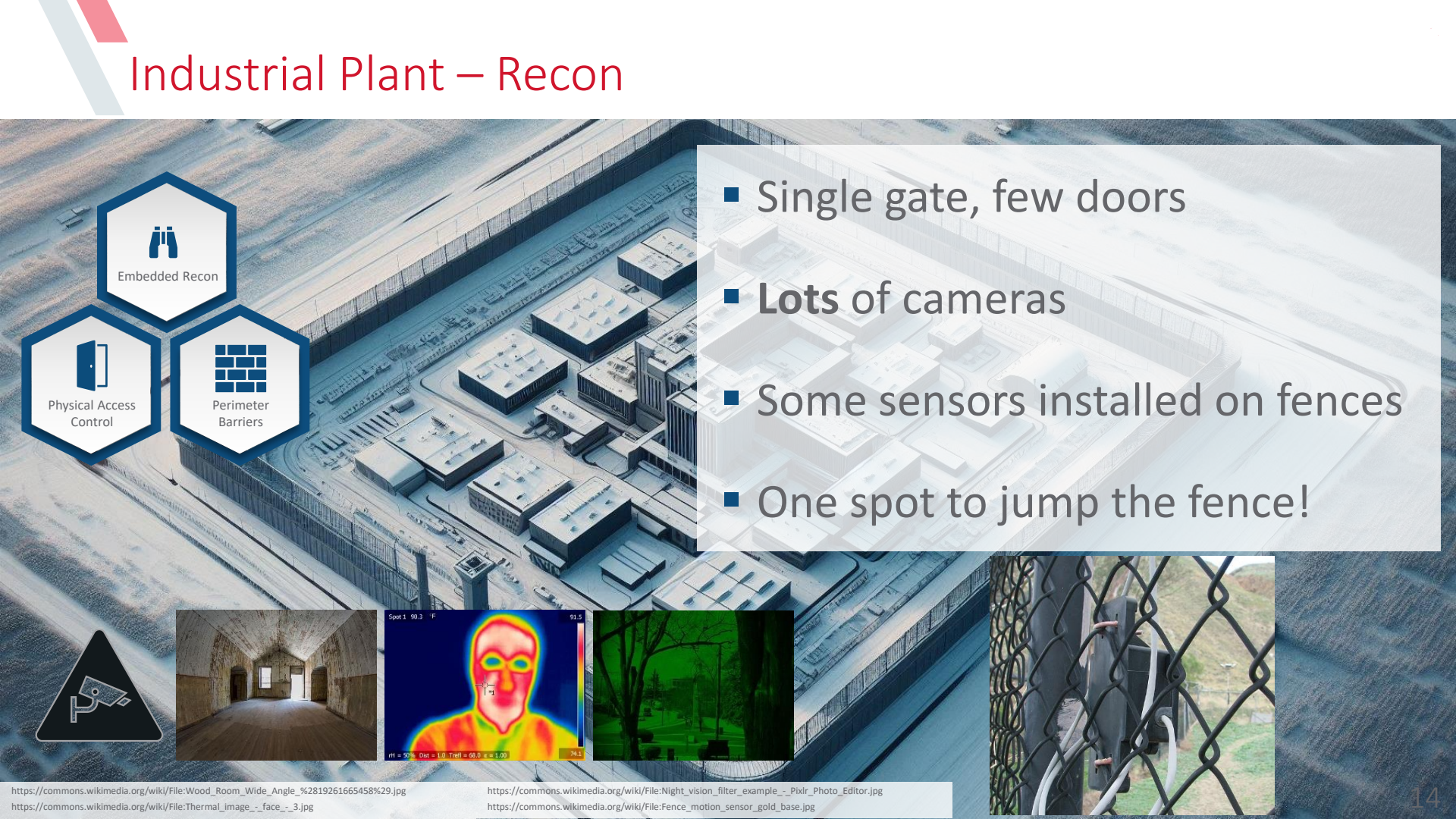
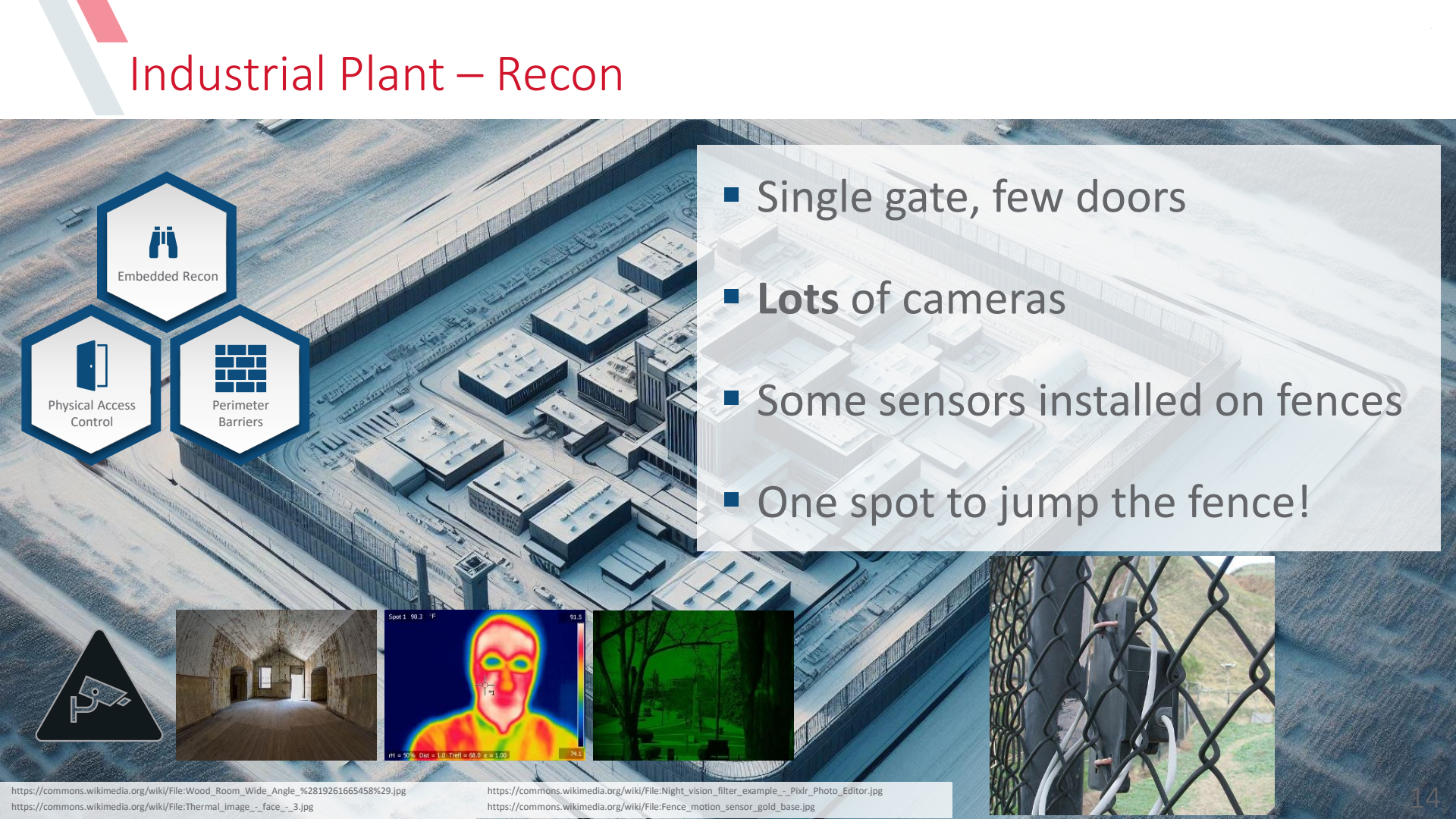
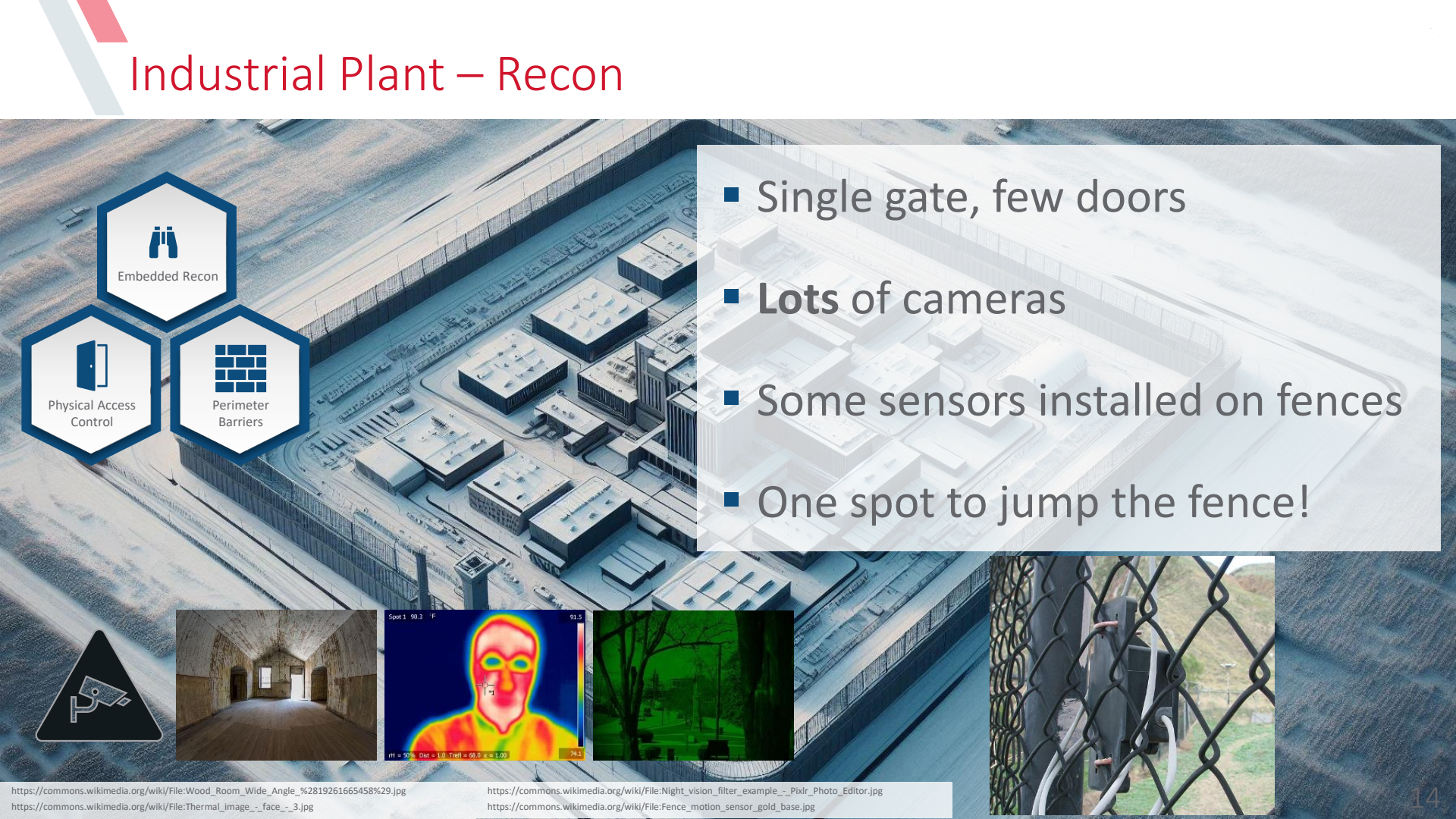
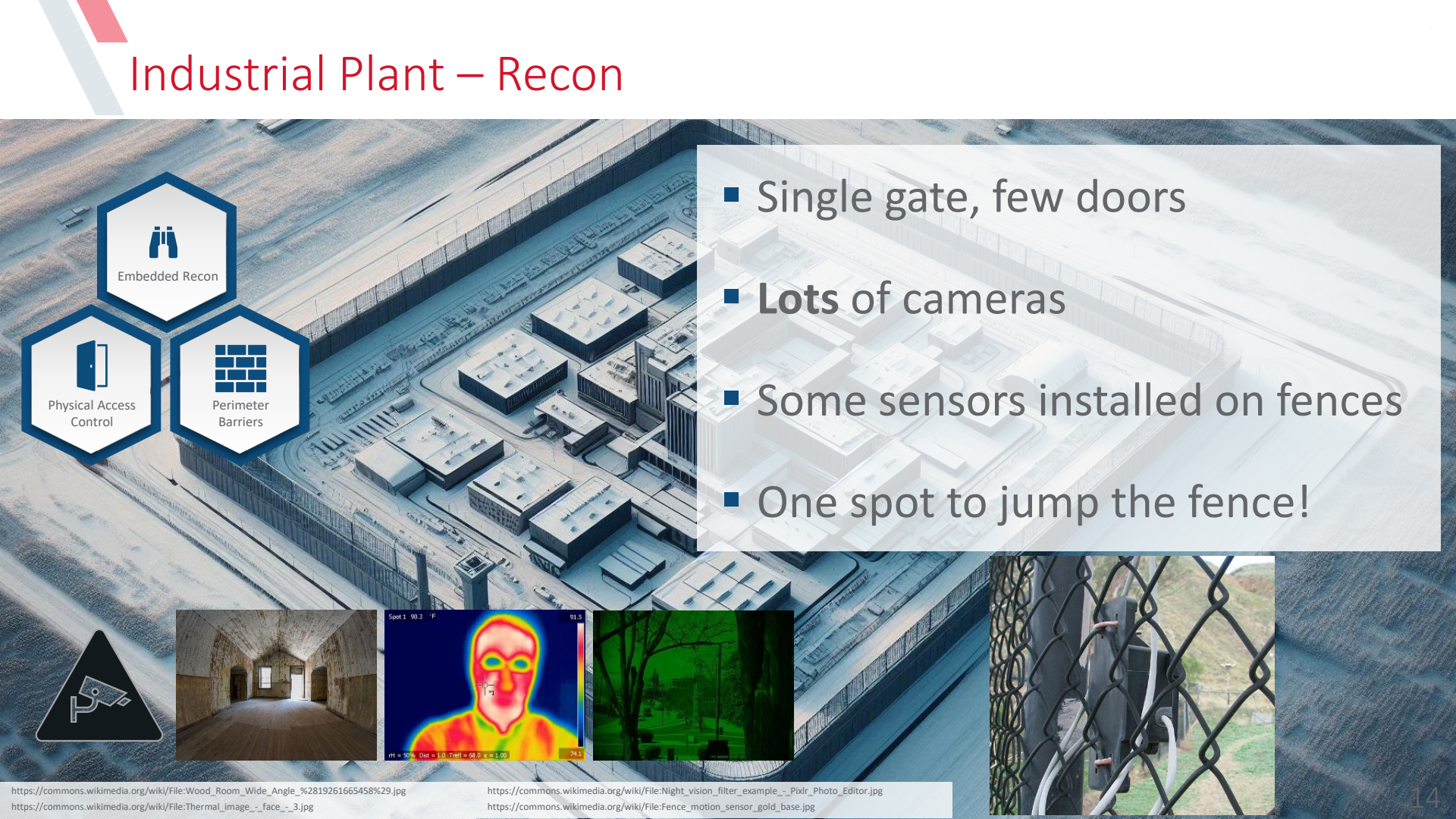
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https://commons.wikimedia.org/wiki/File:Night_vision_filter_example_-_Pbfr_Photo_Editor.jpg

https://commons.wikimedia.org/wiki/File:Fence_motion_sensor_gold_base.jpg

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- # Industrial Plant – Recon
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- Single gate, few doors
 - **Lots** of cameras
 - Some sensors installed on fences
 - One spot to jump the fence!
- Embedded Recon
- Physical Access Control
- Perimeter Barriers
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Industrial Plant – Option #1: Jump the fence

- Easy to get in!
- Hard to get out...
- Suspicious at night
- Can't hide: footprints

WT's Verdict: Won't do.



Industrial Plant – Option #2: Cleaning staff

- Pretext:

- Little preparation
- Simple execution
- Good enough?

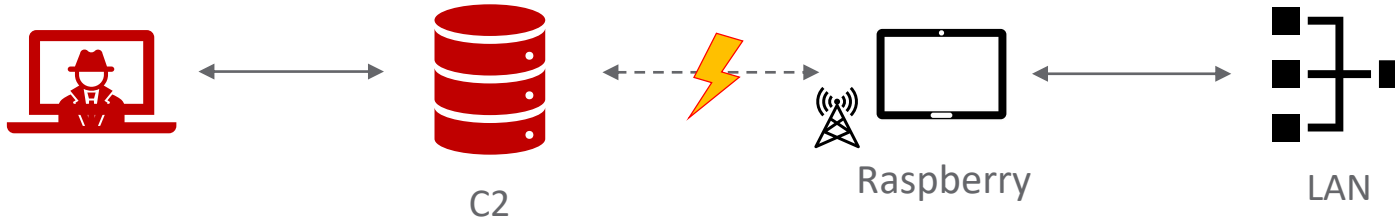
- No keys/keycards.

WT's Verdict: Go for it!



Industrial Plant – Preparation

The evening before action day...

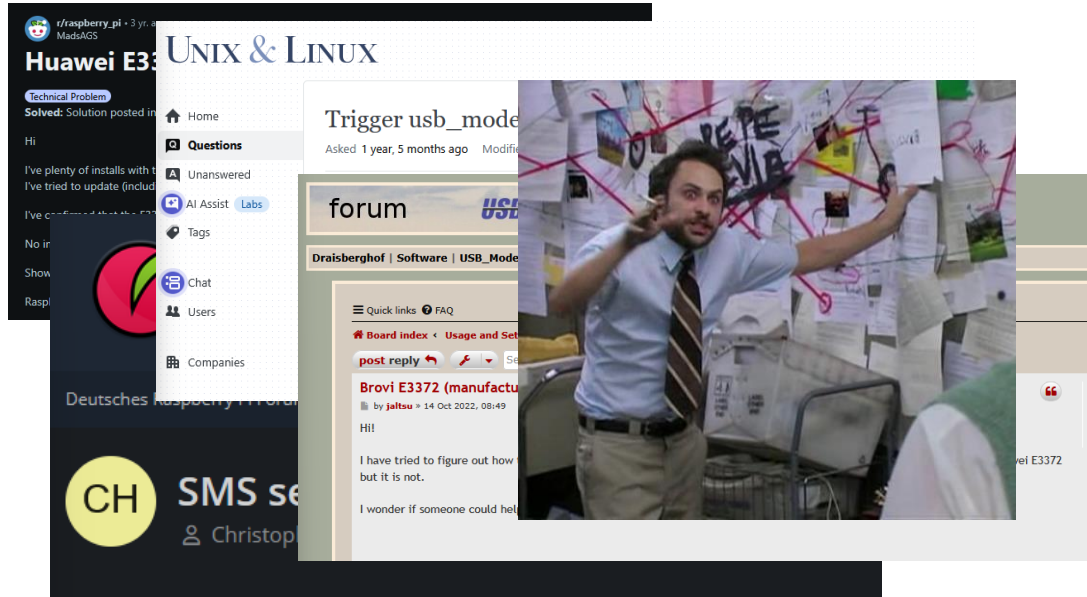


Oh no 🤖

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Industrial Plant – ~~Preparation~~ Oh crap

Linux USB network device troubleshooting speedrun ...until 2am...



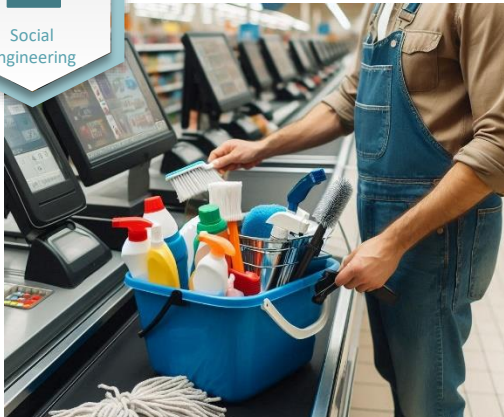
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1 cat /usr/local/bin/stick_startup.sh
2 #!/bin/bash
3
4 echo "##### 2 @reboot bash /usr/local/bin/stick_startup.sh >> /var/log/brovi.log
5 echo " CONFIG
6 echo "#####"
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8 echo "(Sleeping 2min)"
9 sleep 120
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12 usb_modeswitch -v
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When there's some obscure, poorly supported/documented USB periphery, chances are that someone before you tried to make it work on Linux before, horribly failed and eventually succeeded.

Industrial Plant - Entry



Social
Engineering



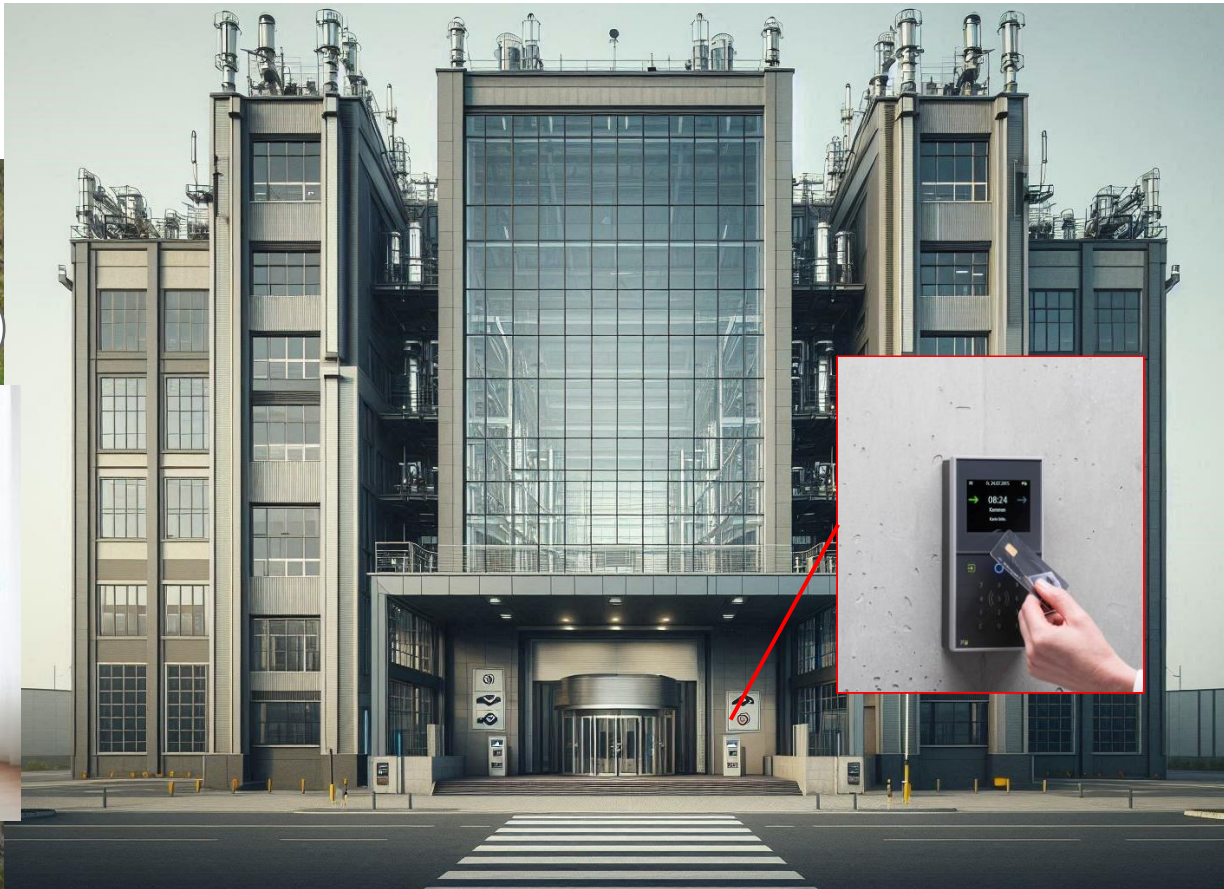
Classification: Internal



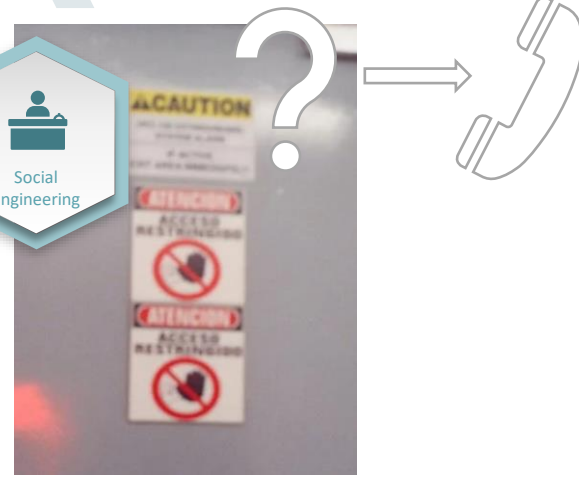
Industrial Plant - Entry



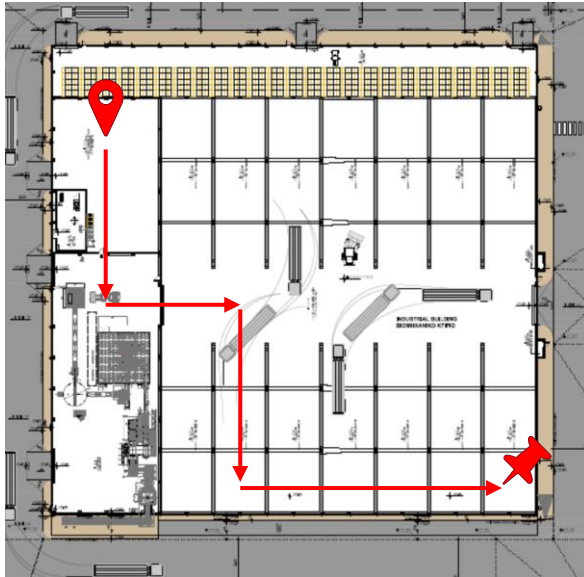
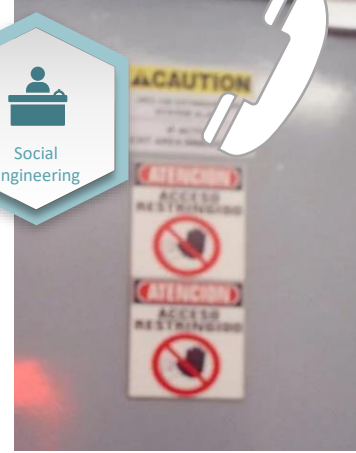
Social
Engineering



Industrial Plant - Entry



Industrial Plant - Entry



Industrial Plant - Entry



Hi, we're the
cleaning staff!

?! 🤖

🕒 1hour later

Industrial Plant – Results



- Social engineering successful
- Bypassed perimeter security
- Lacking security awareness of staff
- Local incident raised
 - Wanted posters



Classification: Internal





Who are you?

Who are you?



Objectives:

- Gain physical access
- Plant rogue devices
- Pictures of sensitive documents



Hardware
Additions



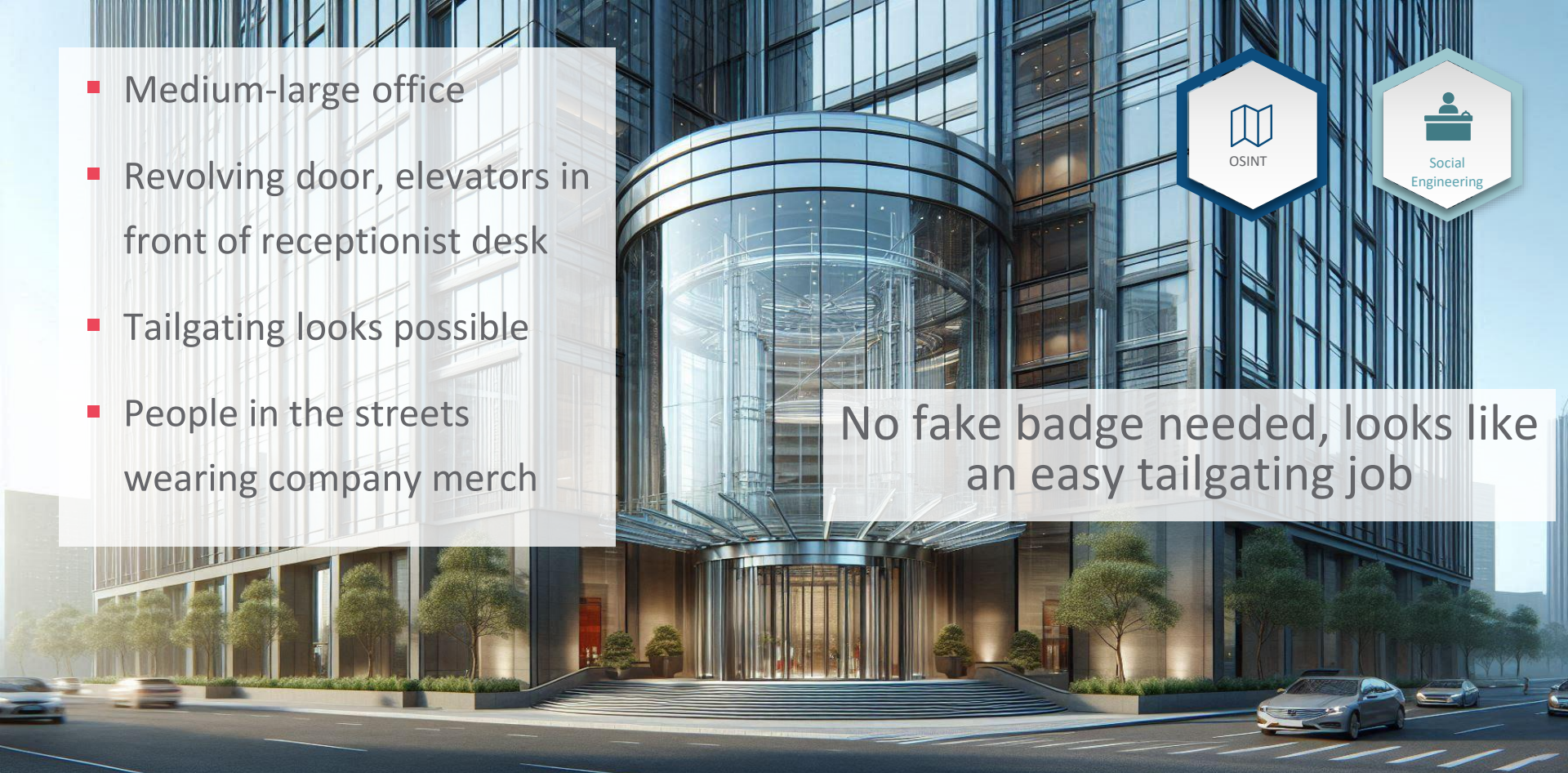
Document
access

Who are you? - Recon

- Medium-large office
- Revolving door, elevators in front of receptionist desk
- Tailgating looks possible
- People in the streets wearing company merch



No fake badge needed, looks like an easy tailgating job



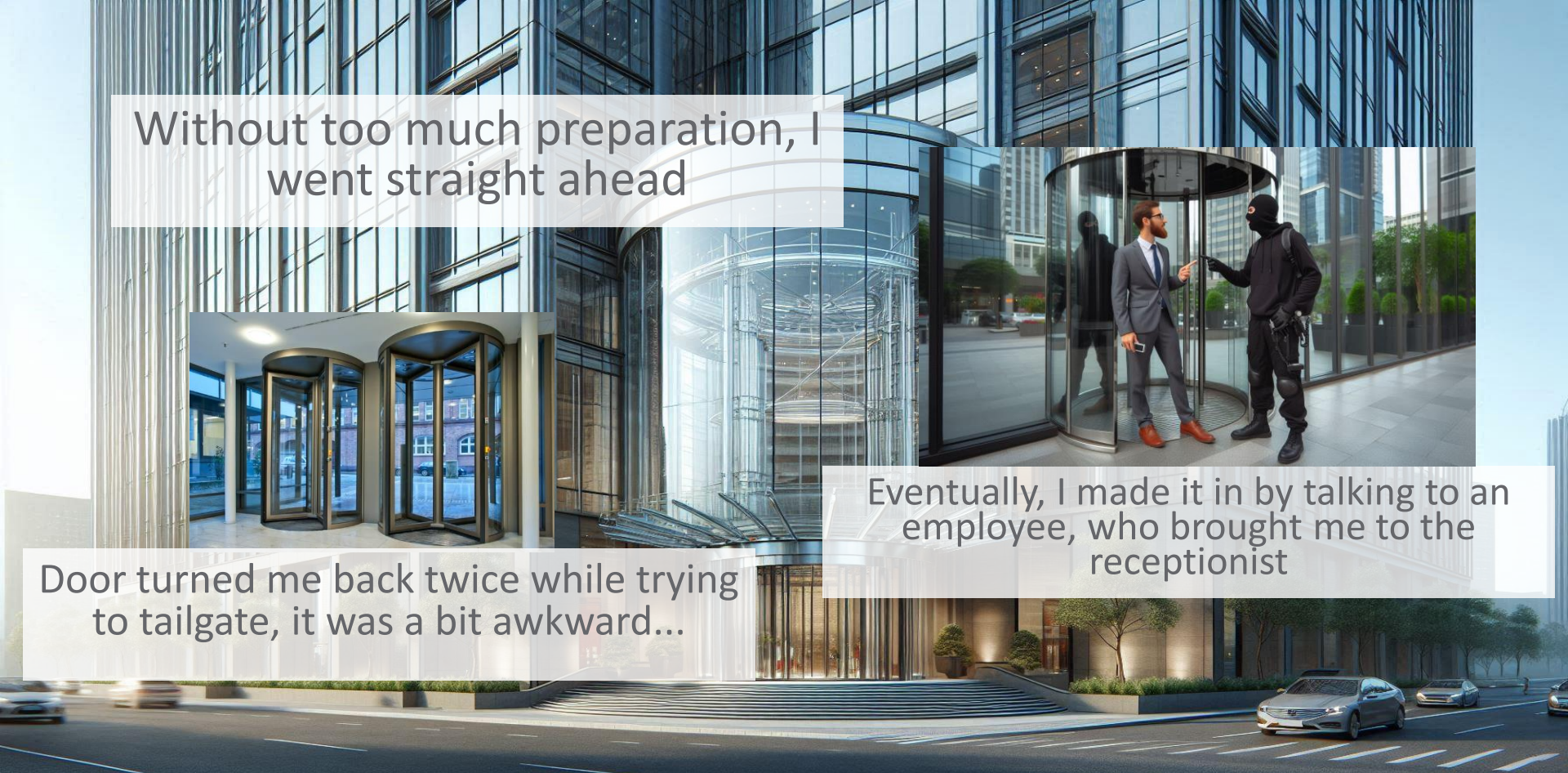
Who are you? - Action

Without too much preparation, I went straight ahead



Door turned me back twice while trying to tailgate, it was a bit awkward...

Eventually, I made it in by talking to an employee, who brought me to the receptionist





Well, at least I am
inside, but stuck
here for the time
being

You can wait
here for your
contact person

It seems all doors and elevators require badges, so how do we get out of here?

No thank you, I'll wait

(3 minutes later)
Hold on, what did she just say??



Actually, I would like to go!

Do you want to eat your food in the canteen?



Badging is normally required to enter the canteen

Those guys are finishing up, I'll follow them into the elevator

Sadly, they went outside the building, which left me in the hall, hiding from the receptionist



A group of seven business professionals (four men and three women) are standing in an elevator. They are all dressed in formal business attire. In the center of the group, a person is wearing a black hooded sweatshirt and a black balaclava, completely obscuring their face. The elevator has dark wood-paneled walls and a control panel with multiple buttons on the right side. The floor is covered with a grey carpet. The lighting is somewhat dim, creating a serious and tense atmosphere.

Eventually, a group arrived that went into the elevators, I quickly snuck inside with them...

I followed them into their office floor and sat down at a desk somewhere

Two guys came up to have a small chat with me, but left afterwards.

Jackpot, let's get this operation going



But look who's back, and they brought a friend... the receptionist lady



A photograph of three people against a plain grey background. On the left, a man with a beard and brown hair wears a brown sweater and has his arms crossed. In the center, a woman with long brown hair wears a blue and white striped shirt and has her hands held out palms up. On the right, another man with a beard and brown hair wears a dark grey sweater and has his arms crossed. All three have serious, questioning expressions. Four speech bubbles are overlaid on the image, containing text that appears to be a script or dialogue.

They started questioning me, sometimes
talking through each other

Who are you? This is a
sensitive department,
only authorized
personnel is allowed!

What are you doing
here? You were
supposed to be
waiting downstairs!

Do you have a contact
person and their
number? I have to call
them.

Somehow, using pure improvisation, I
managed to convince them to let me go...
with a bonus present



Alright, I will guide you
outside. We will get
you a visitor's badge
for another floor.



All is good and well, people are taught to be
helpful. I picked up my new badge and
went back to the office

Sometimes though, silence is golden,
especially when you've already won...





Mountain Base

Mountain Base



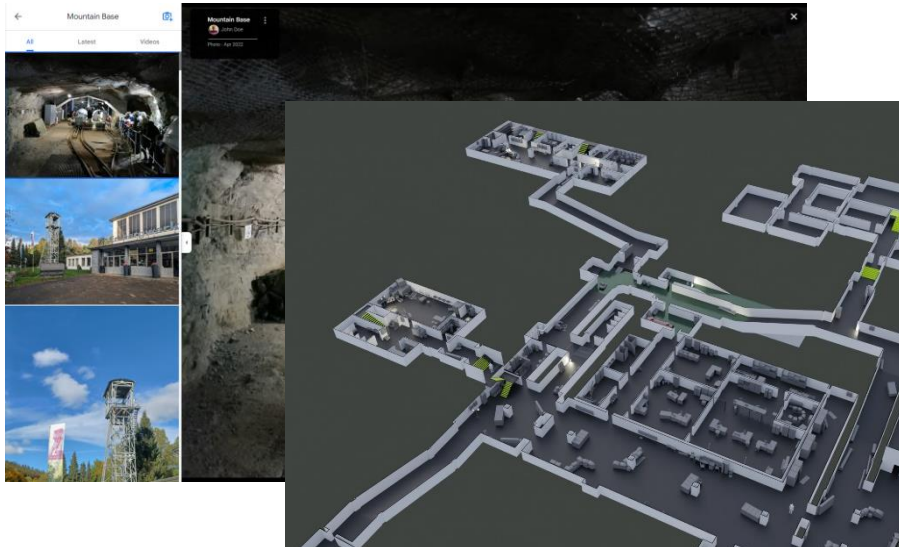
Objectives:

- Gain physical access
- Plant rogue device



Hardware
Additions

Mountain Base – Recon



- Remote location
- Extensive underground area
- Closed to public
 - Guided tours!

Mountain Base – Recon



- High-security locks
- Keypad & Card Reader
- Lots of traffic



Mountain Base – Entry?



Phase II

- Covert Entry
- Overt Entry



Classification: Internal

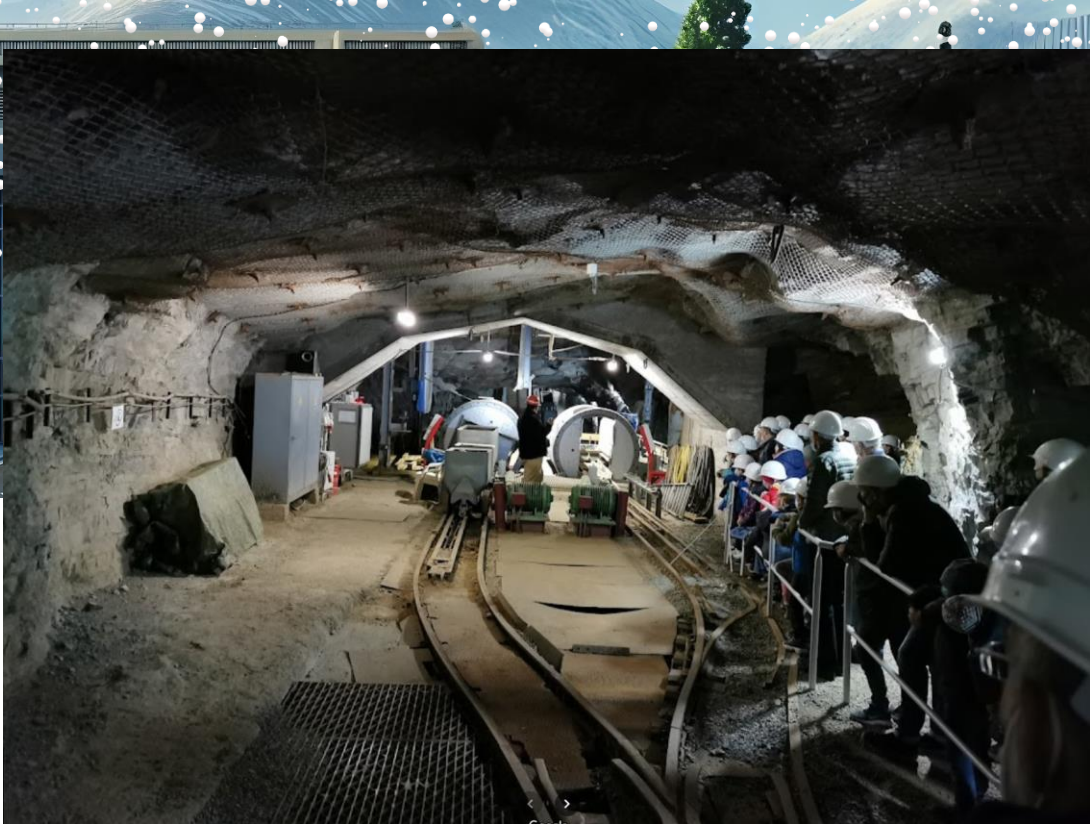
Mountain Base – Entry I



Social
Engineering



Persistence



What's not together?



Mountain Base – Entry I - Results



- Process oriented
 - Deviations spark suspicion
- Persistent & Assertive
- Internal incident raised at HQ



Mountain Base – Entry II

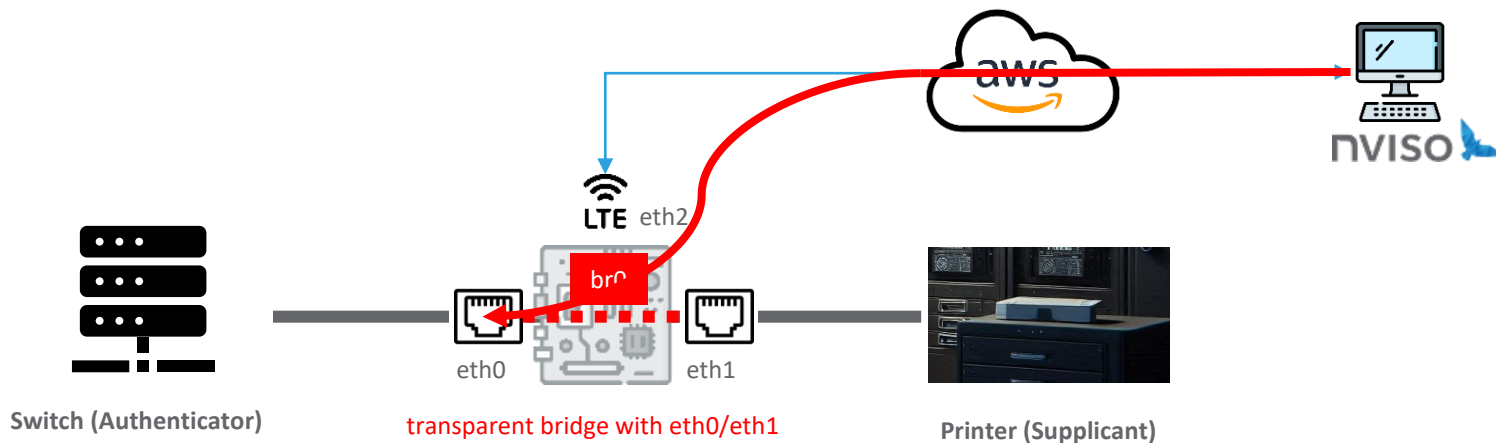


Alarm and CCTV
deactivation



Mountain Base – Entry II

Hardware Implant – NAC Bypass



Packet interaction configuration:

- internal routing with static ARP entry to 00:aa:bb:cc:dd:ee:ff
- source NAT for 1.2.3.4
- source NAT for 00:11:22:33:44:55:66

Mountain Base – Entry II



Alarm and CCTV
deactivation



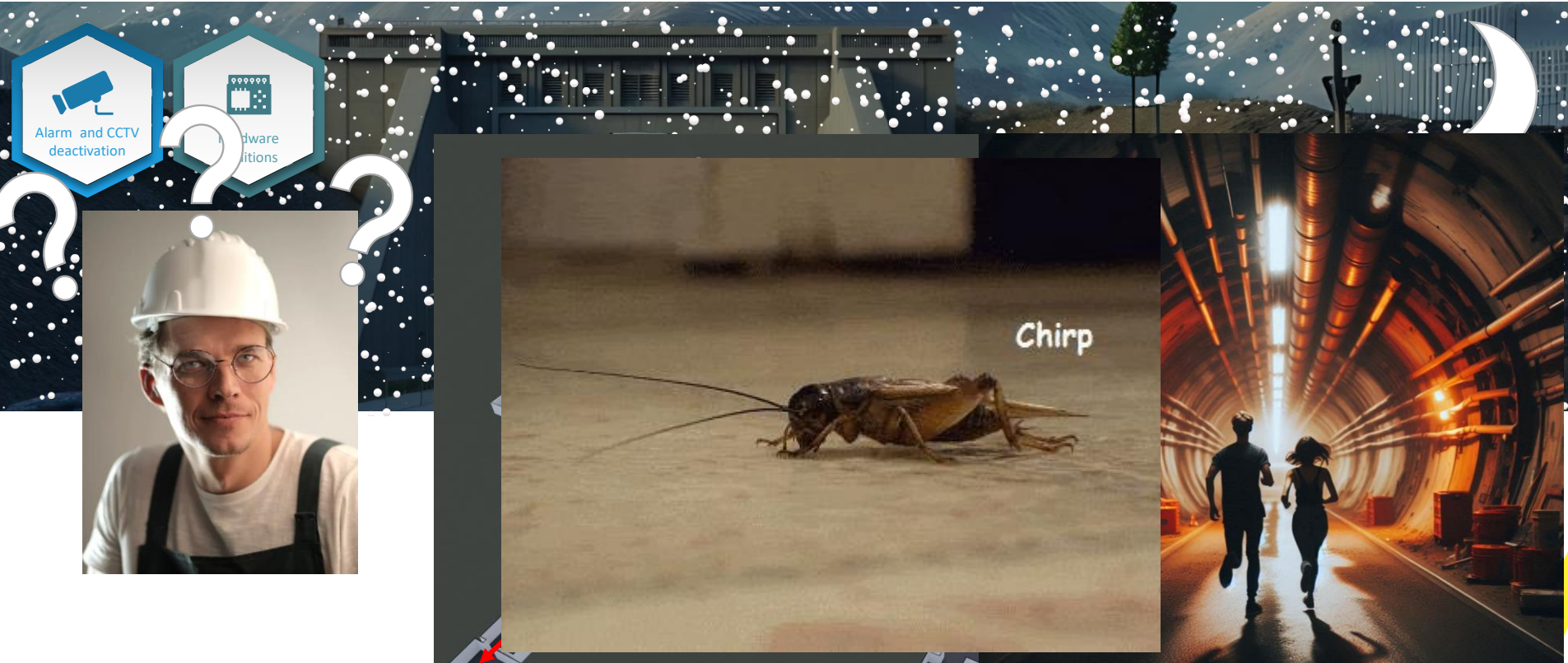
Hardware
Additions



If I were you...
I'd **RUN!**



Mountain Base – Entry II



Mountain Base – Entry II - Results



- Leg-up successful
- Heavy security at entrance only
- **Local incident raised**
 - No connection to first incident
 - Plant-wide investigation & search



Classification: Internal



Vocal Cyber Threat

Vocal Cyber Threat



Objectives:

- Gain physical access
- Plant rogue devices
- Pictures of sensitive documents



Hardware
Additions



Document
access

Vocal Cyber Threat - Recon



OSINT



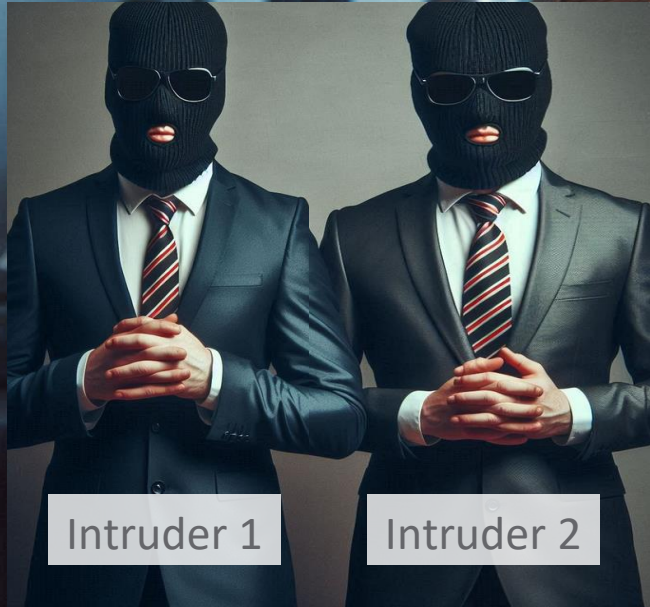
Access card
cloning

- Badge cloning is possible, but takes a while

- Very big office
- Entry only via receptionist or badge
- Tailgating not possible

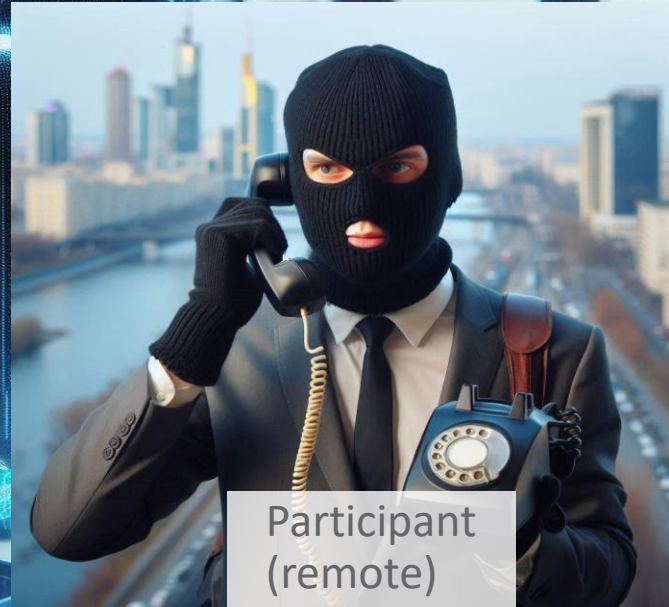
Vocal Cyber Threat - Entry

The attack plan is voice phishing, which can be incredibly effective when done correctly



Intruder 1

Intruder 2



Participant
(remote)

Vocal Cyber Threat - Preparations

To perform vishing, we will need some phone numbers to call

How can we get phone numbers of employees?



The dark web



Virtual Breach



Out-of-office mails

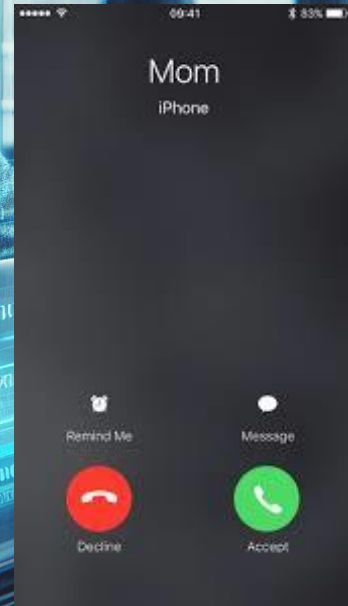
Vocal Cyber Threat - Preparations

To make it even more effective, we apply caller ID spoofing

Call someone and display any number

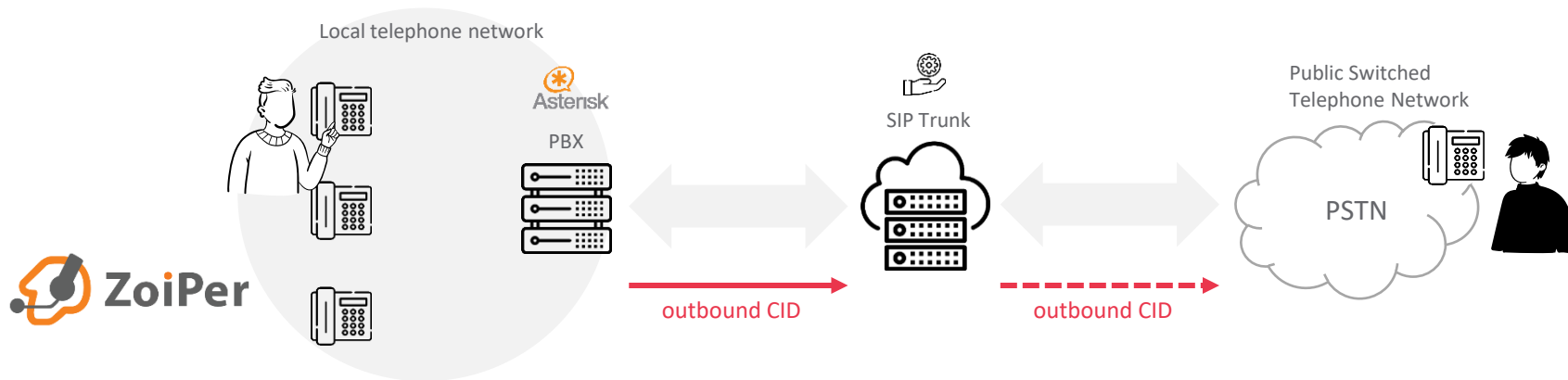


If the number is saved in the target's contacts, it will display the contact name instead



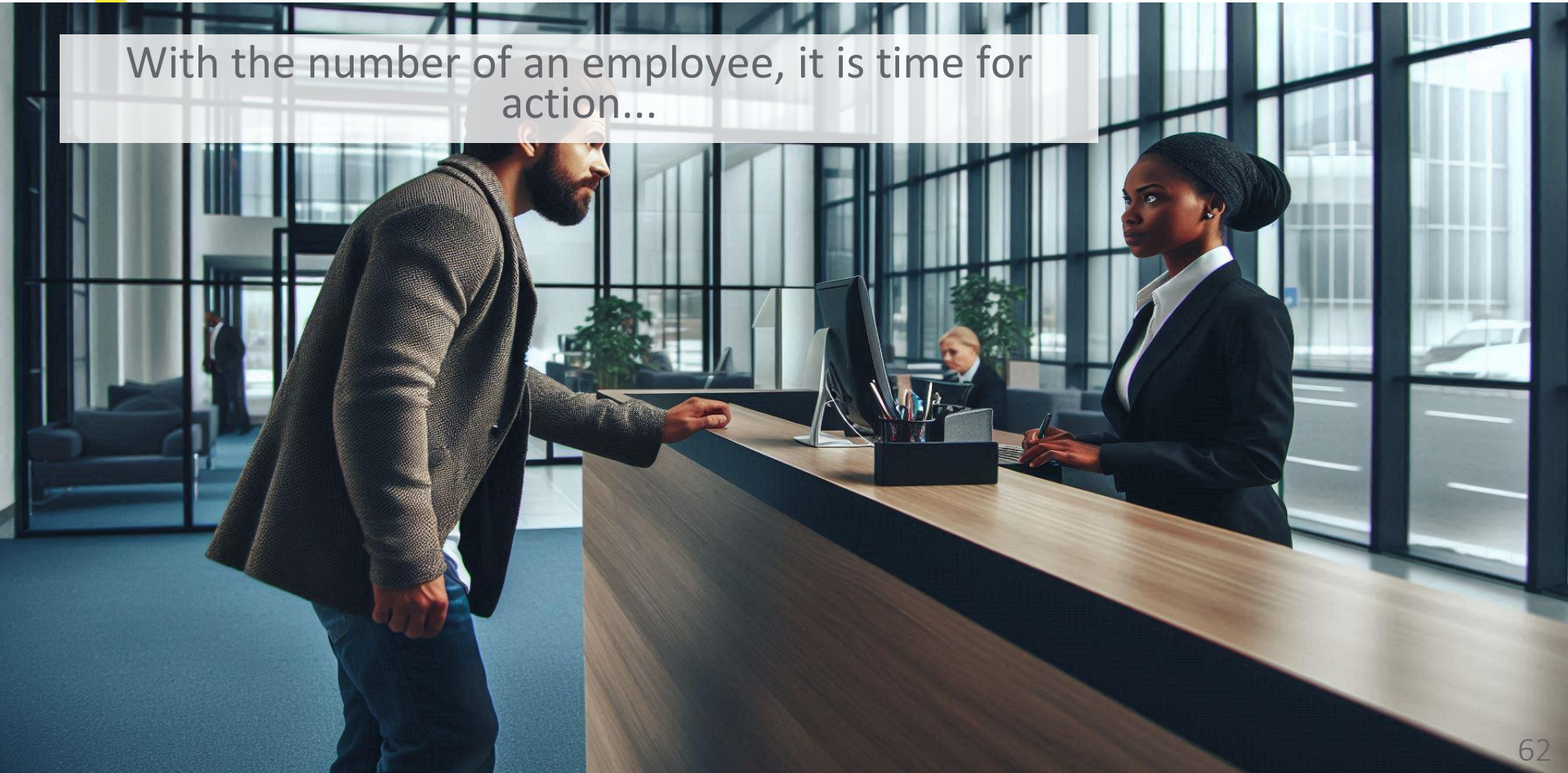
Spoofing Caller-ID

SIP Trunking



Vocal Cyber Threat - Action

With the number of an employee, it is time for action...





Hey sorry, my battery is dead and I urgently need to call my colleague. Could you call them for me, please?

Of course, could you tell me their number?

A man with a beard and a grey blazer is leaning over a wooden reception desk, talking to a female receptionist. The receptionist is wearing a black blazer and has her hair in a bun. They are in a modern office with large windows and a blue carpet. A speech bubble from the man says, "Sure, it is <participant's number> .". A speech bubble from the receptionist says, "Thank you, I'll call them right now.".

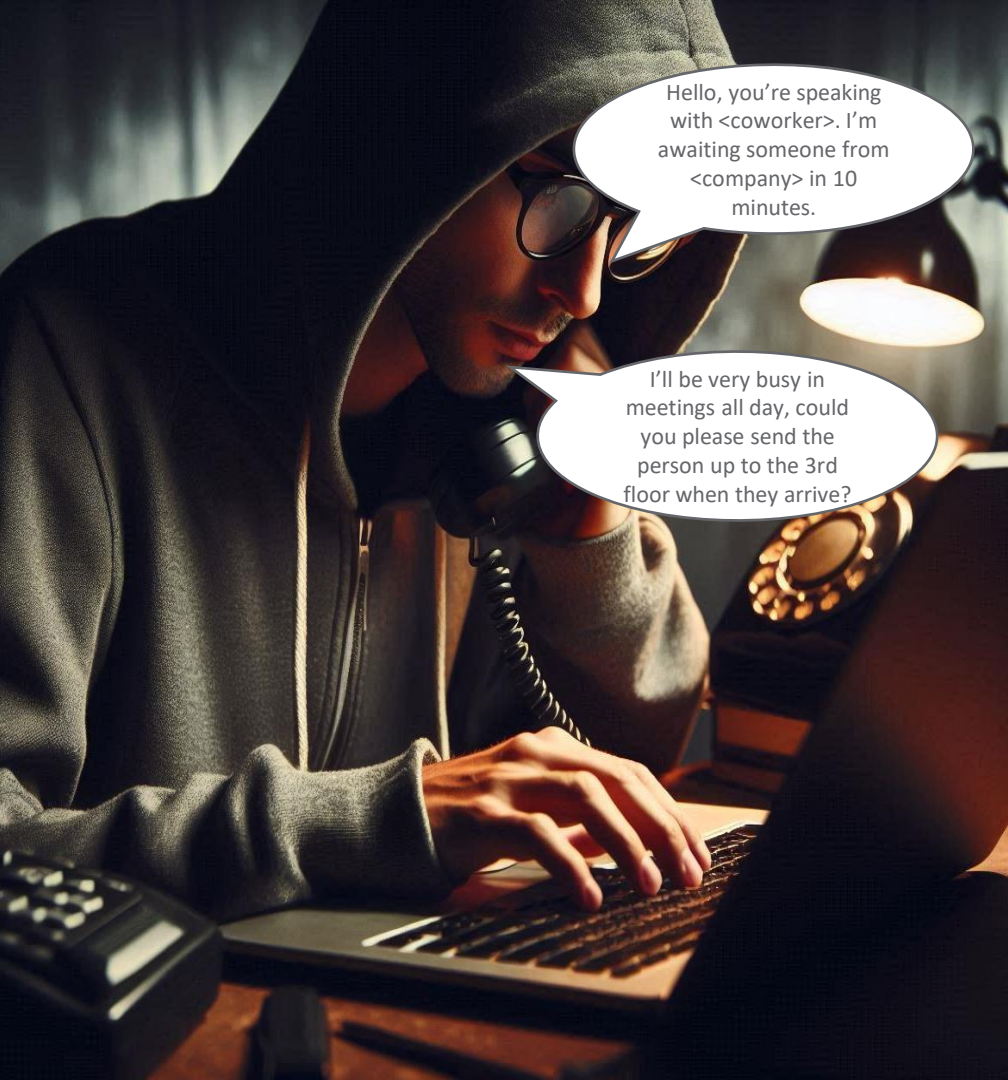
Sure, it is <participant's number> .

Thank you, I'll call them right now.

The receptionist calls our remote participant and now we have the receptionist's number

A person wearing a grey hoodie and black-rimmed glasses is sitting at a desk in a dimly lit room. They are holding a black telephone receiver to their ear with their left hand and typing on a laptop keyboard with their right hand. A desk lamp with a warm, yellowish glow is positioned to the right, illuminating the person's face and the desk. In the foreground, a black multi-line office phone is visible on the left. The background is dark and out of focus, suggesting a window with curtains.

Our participant uses caller ID spoofing to call the receptionist using a phone number of one of their employees (AI generated voice is also an option)



Hello, you're speaking with <coworker>. I'm awaiting someone from <company> in 10 minutes.

I'll be very busy in meetings all day, could you please send the person up to the 3rd floor when they arrive?



I've noted their name and company. I'll send them straight to you!

A man with a beard, wearing a grey and black striped sweater, stands on the left side of the frame, gesturing with his hands as he speaks. He is facing a woman who is seated at a wooden desk on the right. The woman has dark hair pulled back and is wearing a dark blazer over a white collared shirt. She is looking towards the man. The background consists of large glass windows with black frames, showing a view of a city street with buildings and a car. A small potted plant sits on the desk between them.

Hello, sorry that I am a bit late. My name is <name> from <company>.

Ah yes, I was told by <coworker> that you'd arrive soon. I'll open the door for you.

Our second intruder comes in as the “awaited guest”

A conceptual illustration of badge cloning. A large hand is shown holding a blue badge with the text 'ENTRANCE' and '000100' and inserting it into a cloning device. The device is a small, white, rectangular unit with a black base and several cables connected to it. A miniature figure of a man in a suit, carrying a briefcase, is walking on the device. In the foreground, there are several other badges and a calculator. One badge is labeled 'ENTRANCE' and another '000100'. A calculator is visible in the background. The scene is set on a light-colored wooden surface.

Optionally, intruder 2 could now use a badge cloning device to find badges left unattended by workers to let in intruder 1

Badge cloning



Mifare Classic
(13.56 MHz)



~ 30 seconds



iCopy-X

Copy



Mifare cloning card

default encryption key set (0xFF FF FF FF FF FF)
CRYPTO1 stream cipher -> weak, predictable encryption
algorithm

Takeaways



- Trust but verify: be persistent & assertive



- Ensure that visitors are always accompanied by staff



- Always verify a visitor's contact person before letting them in



- Verify employee phone numbers before calling them, do not call numbers that are given to you by strangers



Thank you!



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